SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1220	**Course Title:**	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	**Assessment:**	Industry Problem solving task
Weightage:	30%	**Total Marks:**	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	**Slot-B / 2025**	**Date:**	**18.08.2026**

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A mobile device manufacturer observes noticeable lag when users switch rapidly between multiple applications. Analyze how L1, L2, and L3 caches and DRAM participate in storing and retrieving frequently accessed application data. Based on latency, capacity, hit rate, and bandwidth requirements, propose a suitable hierarchical memory organization that can reduce application switching delays and improve overall system throughput.
2. A cloud service provider experiences increased latency when processing database queries from a large number of concurrent users. Analyze how cache hierarchy and virtual memory paging influence memory access performance, and compare their roles in reducing processor-memory latency. Recommend suitable memory technologies and memory-management improvements that can minimize page faults, reduce access delays, and improve database throughput.
3. An autonomous vehicle continuously receives large volumes of sensor data from cameras, LiDAR, radar, and other real-time sensing systems. Evaluate the characteristics of SRAM, DRAM, and MRAM in terms of access latency, bandwidth, capacity, volatility, and power consumption. Based on the real-time processing requirements, design an appropriate memory hierarchy that can provide high data-transfer bandwidth while minimizing access latency and supporting reliable sensor-data processing.
4. An edge-based AI device must load machine-learning models quickly while operating under strict power and storage constraints. Analyze the characteristics of NAND Flash, DRAM, and RRAM with respect to access speed, storage capacity, persistence, power consumption, and suitability for AI workloads. Recommend a hierarchical memory organization that enables fast model loading, efficient intermediate-data processing, reduced energy consumption, and improved overall inference performance.
5. A high-performance gaming console experiences frame drops and noticeable delays when loading large game environments containing high-resolution textures, audio, and other assets. Analyze the

interaction between L3 cache and DRAM during the retrieval and processing of frequently accessed game data, considering their latency, capacity, and bandwidth. Propose suitable memory hierarchy enhancements that can reduce data-access bottlenecks, increase memory throughput, and provide smoother frame rendering during large scene transitions.

Code of Conduct

I Vishwa.U(192511192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understandi ng of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understandin g; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentati on and presentation	Basic documentati on with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Applicati on of Engineeri ng Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professio na lism(5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/5	/ 100

Course Coordinator Faculty Incharge

Q1. Mobile Application Switching – Cache Hierarchy

Industry Problem Analysis:
When a smartphone user rapidly switches between applications, the processor repeatedly accesses application instructions, UI data, recently used variables, and operating-system data. If this data is not available in the faster cache levels, the processor must access DRAM, increasing memory-access latency and causing application-switching delays.

Engineering Concept Application:

- **L1 Cache:** Smallest and fastest cache, located close to the CPU core. It stores the most frequently and recently accessed instructions and data.
- **L2 Cache:** Larger than L1 but slightly slower. It stores data that cannot fit in L1 and provides a second-level backup.
- **L3 Cache:** Larger shared cache that reduces the number of accesses to DRAM.
- **DRAM:** Main memory with much higher capacity but significantly greater latency than cache.

A suitable hierarchy is:
CPU → L1 → L2 → L3 → DRAM → Flash Storage
Frequently accessed application data should remain in L1/L2, while shared application and OS data can be maintained in L3. Less frequently accessed data can remain in DRAM.

Performance Analysis:
The average memory access time can be represented as:
For example, assume:

- L1 hit rate = 95%
- L2 hit rate for L1 misses = 90%
- L3 hit rate for remaining misses = 80%

For 100 memory requests:

- 95 requests are satisfied by L1.
- 5 requests miss L1.
- 4.5 requests are satisfied by L2.
- 0.5 requests reach L3.

- 0.4 requests are satisfied by L3.
- Only 0.1 requests reach DRAM.

Thus, hierarchical caching significantly reduces expensive DRAM accesses.

Proposed Solution:
Use a multi-level cache hierarchy with high L1/L2 hit rates, an adequately sized shared L3 cache, efficient prefetching, and operating-system memory management. Application data should be organized to improve **temporal and spatial locality**.

Expected Outcome:
The proposed hierarchy reduces DRAM accesses, decreases application-switching latency, improves CPU utilization, and increases smartphone system throughput.

Conclusion:
A properly optimized **L1–L2–L3–DRAM hierarchy** is suitable for smartphones because it balances latency, capacity, bandwidth, and power consumption.

Q2. Cloud Database Server – Cache and Virtual Memory

Industry Problem Analysis:
A cloud database server handling thousands of concurrent users requires rapid access to frequently used indexes, query data, instructions, and database buffers. Excessive cache misses and virtual-memory page faults can significantly increase query latency.

Engineering Concept Application:
The memory hierarchy can be represented as:

CPU → L1 → L2 → L3 → DRAM → SSD/NAND

Cache memory reduces processor-memory latency by keeping frequently accessed data close to the CPU.

Virtual memory allows applications to use an address space larger than physical DRAM. Memory is divided into **pages**, and pages not currently available in RAM may have to be retrieved from secondary storage.

A **page fault** occurs when the required page is not present in physical memory.

Cache vs Virtual Memory:

Feature	Cache	Virtual Memory
Main purpose	Reduce CPU-memory latency	Extend apparent memory capacity
Location	Between CPU and DRAM	DRAM + secondary storage
Speed	Very high	Much slower during page faults
Unit	Cache block	Page
Main problem	Cache miss	Page fault

Performance Analysis:

Suppose:

- Normal DRAM access = 100 ns
- Page-fault service time = 5 ms
- Page-fault rate = 0.01%

Average access time:

Therefore, even a small page-fault rate can significantly increase average memory-access time.

Proposed Solution:

1. Increase DRAM capacity.
2. Maintain frequently accessed database pages in memory.
3. Use large and efficient database buffer pools.
4. Optimize L1/L2/L3 cache locality.
5. Reduce unnecessary memory allocation.
6. Use SSD/NAND storage for faster page-fault recovery.
7. Apply appropriate page-replacement policies.
8. Reduce random memory access patterns.
9. Use memory-aware database indexing.

Expected Outcome:

Fewer cache misses and page faults reduce CPU waiting time and improve database query throughput.

Conclusion:

Cache hierarchy primarily addresses **processor-memory latency**, whereas virtual memory manages **capacity and address-space requirements**. A combination of large DRAM, efficient cache utilization, optimized paging, and fast SSD storage provides a suitable cloud-server memory architecture.

Q3. Autonomous Vehicle – SRAM, DRAM and MRAM

Industry Problem Analysis:

An autonomous vehicle continuously receives large amounts of data from cameras, LiDAR, radar, ultrasonic sensors, and other systems. The memory system must process this data with very low latency while providing high bandwidth and reliable operation.

Technology Comparison:

Characteristic	SRAM	DRAM	MRAM
Latency	Very low	Low–moderate	Low
Bandwidth	Very high	High	High
Capacity	Low	High	Moderate
Volatility	Volatile	Volatile	Non-volatile
Refresh required	No	Yes	No
Power	High per bit, but fast	Requires refresh	Low standby
Main application	Cache/buffers	Main memory	Fast persistent memory

Proposed Memory Hierarchy:

Sensor → SRAM Buffer → L1/L2/L3 Cache → DRAM → MRAM/Storage

- **SRAM:** Used for immediate sensor buffering and critical real-time processing.
- **DRAM:** Used for large camera frames, LiDAR point clouds, radar data, and AI intermediate data.
- **MRAM:** Suitable for persistent configuration, critical system information, logs, and selected fast-access data.

Performance Analysis:

Suppose the system receives:

- Camera data = 2 GB/s
- LiDAR data = 1 GB/s
- Radar and other sensors = 0.5 GB/s

Total incoming bandwidth:

The memory system should therefore provide significantly more than **3.5 GB/s** of effective bandwidth to accommodate processing, buffering, and simultaneous memory operations.

Proposed Solution:

Use SRAM for latency-sensitive sensor buffers, high-bandwidth DRAM for large-scale sensor processing, and MRAM for non-volatile critical information. DMA can be used to transfer sensor data directly to memory while reducing CPU overhead.

Expected Outcome:

The architecture provides high bandwidth, reduces processing latency, minimizes CPU waiting, and improves reliable real-time sensor processing.

Conclusion:

A combination of **SRAM + DRAM + MRAM** provides an effective memory hierarchy for autonomous vehicles because each technology is assigned according to latency, capacity, persistence, and power requirements.

Q4. Edge AI Device – NAND Flash, DRAM and RRAM

Industry Problem Analysis:

Edge AI devices need to load machine-learning models quickly while operating with limited power, memory, and storage. AI inference also requires rapid access to intermediate tensors, weights, and activation data.

Technology Comparison:

Characteristic	NAND Flash	DRAM	RRAM
Volatility	Non-volatile	Volatile	Non-volatile
Capacity	High	High	Moderate
Access speed	Slower	Very fast	Fast
Persistence	Yes	No	Yes
Power	Low standby	Higher	Low
AI suitability	Model storage	Runtime processing	Fast/persistent AI data

Proposed Memory Hierarchy:

NAND Flash → RRAM → DRAM → CPU/NPU Cache

- **NAND Flash:** Stores large ML models and application data permanently.
- **RRAM:** Stores frequently used model parameters or data requiring persistence with faster access.
- **DRAM:** Stores active model weights, input data, activations, and intermediate computation results.
- **Cache:** Keeps frequently used AI data close to the processor/NPU.

Performance Analysis:

Suppose a 1 GB model is loaded from storage.

If effective storage bandwidth is:

Then approximate loading time is:

If the effective bandwidth is increased to:

then:

Thus, improving the memory path can reduce model-loading time by approximately **50%**.

Proposed Solution:

1. Store the complete model in NAND Flash.
2. Keep frequently used model components in RRAM.
3. Load active models into DRAM.
4. Use processor/NPU cache for frequently reused tensors.
5. Apply model quantization and compression.
6. Use DMA for efficient model transfer.
7. Minimize unnecessary data movement between memory levels.

Expected Outcome:

The architecture reduces model-loading time, decreases energy consumption, improves data locality, and increases AI inference performance.

Conclusion:

A **NAND Flash → RRAM → DRAM → Cache** hierarchy provides a good balance between persistence, capacity, latency, power, and AI-processing requirements.

Q5. Gaming Console – L3 Cache and DRAM

Industry Problem Analysis:
Modern games continuously access large textures, 3D models, audio, shaders, physics data, and world information. During large scene transitions, insufficient memory bandwidth or excessive cache misses can cause frame drops and rendering delays.

Engineering Concept Application:

The memory hierarchy is:

CPU/GPU → L1/L2 → L3 Cache → DRAM → SSD

The **L3 cache** stores frequently reused game data close to the processor. When required data is not available in L3, the system accesses DRAM.

DRAM has much greater capacity than L3 but higher latency.

Data Locality:

- **Temporal locality:** Frequently reused textures, game objects, and instructions remain in cache.
- **Spatial locality:** Data stored close together can be fetched efficiently as cache blocks.

Performance Analysis:

Assume:

- L3 hit rate = 90%
- L3 access time = 20 ns
- DRAM access time = 100 ns

Average access time:

If the L3 hit rate decreases to 70%:

Therefore, increasing the L3 hit rate from **70% to 90%** reduces the calculated average access time from **50 ns to 30 ns**, representing a substantial improvement.

Proposed Solution:

1. Increase effective L3 cache capacity.
2. Improve game-data locality.
3. Use intelligent prefetching.
4. Stream large assets from high-bandwidth storage.
5. Organize textures and game assets efficiently.

6. Reduce unnecessary data movement.
7. Provide sufficient DRAM bandwidth.
8. Use parallel memory channels where supported.

Expected Outcome:
Higher cache hit rates reduce DRAM accesses, while sufficient DRAM bandwidth supports large asset transfers. This reduces memory bottlenecks and helps maintain stable frame rates.

Conclusion:
An optimized **L3 cache + high-bandwidth DRAM + fast storage hierarchy** can reduce scene-transition delays and provide smoother gaming performance.

Overall Industry-Level Comparison

Application	Primary Memory Requirement	Recommended Hierarchy
Smartphone	Low latency + power efficiency	**L1 → L2 → L3 → DRAM → Flash**
Cloud Server	High capacity + low latency	**L1/L2/L3 → Large DRAM → SSD**
Autonomous Vehicle	Real-time + high bandwidth	**SRAM → Cache → DRAM → MRAM/Storage**
Edge AI	Fast model loading + low power	**NAND → RRAM → DRAM → Cache**
Gaming Console	High bandwidth + low latency	**L1/L2 → L3 → High-bandwidth DRAM → SSD**

Final Performance Principles

Across all five industry problems, the most important optimization principles are:

1. **Increase cache hit rate** to reduce expensive lower-level memory accesses.
2. **Exploit temporal and spatial locality.**
3. **Use DRAM for high-capacity active workloads.**
4. **Use SRAM where extremely low latency is required.**
5. **Use MRAM/RRAM where persistence and low-power operation are important.**
6. **Use NAND Flash for high-capacity persistent storage.**
7. **Minimize virtual-memory page faults** in memory-intensive systems.
8. **Increase effective memory bandwidth** for sensor, AI, and graphics workloads.
9. **Reduce unnecessary data movement** between hierarchy levels.
10. Design the hierarchy according to the application's **latency, bandwidth, capacity, throughput, and power constraints**.